T3 reduced-scalar numerical benchmarks

Scope: 16 representations, two frozen-UV scenarios; additional scalar couplings initially zero. Historical UV-running inputs are examined as exploratory RGE diagnostics.
Acceptance: Five-observable Gaussianised NuFIT $\chi^2 \leq 5.86$. Individual fits: 32/32. Unchanged scenario-specific A(-2) Yukawas: 10/32.

Frozen-UV Yukawa comparisons

Model	Low ind. χ^2	Low A(-2) χ^2	High ind. χ^2	High A(-2) χ^2
A_+0	1.12149e-11	0.00817089	0.00516295	0.00758579
A_-2	5.30105e-17	5.30105e-17	3.53986e-11	3.53986e-11
A_-4	2.08195e-16	11527.7	2.86028e-14	11464.1
B_+1	1.1989e-21	1.4739	3.44636e-11	2.1874
B_-1	8.93793e-18	37.3954	1.06771e-08	35.4519
B_-3	8.29795e-15	12177.8	1.48272e-09	12124.1
C_+1	6.60223e-20	9273.32	3.48456e-20	9256.32
C_-1	2.28605e-10	9749.82	2.45329e-18	9739.22
C_-3	8.04078e-11	17818.8	8.57308e-15	17797.4
D_+0	2.57559e-12	0.000202375	1.04657e-08	0.000209023
D_+2	4.12536e-12	232.824	8.5912e-11	238.68
D_-2	1.2035e-10	0.0128843	3.4551e-11	0.0122503
E_+0	1.36835e-09	3922.1	4.12798e-11	4106.15
E_+2	2.43592e-17	2932.51	5.58612e-11	3106.76
E_-2	1.18552e-09	3915.55	3.02698e-15	4099.54
E_-4	8.27006e-12	15394.6	8.65198e-11	15437.3

Class E uses $\lambda_{T3} = 2.5 \times 10^{-9}$ (low) and 0.25 (high); classes A–D use 5×10^{-9} and 0.5.

Exploratory fixed-input UV-running diagnostic

The nominal 1 TeV scalar-EFT endpoints below are *formal diagnostics*, not rematched threshold masses. No historical seed satisfies the original 0.5–5 TeV scalar window.

Model	Formal m_{S_1} [TeV]	Formal m_{S_2} [TeV]	ρ_F
A_+0	40.9155	42.3856	0.411017
A_-2	39.89	40.5393	0.380523
A_-4	35.1197	59.4371	0.524996
B_+1	30.4184	37.8432	0.35929
B_-1	72.8071	100.076	1.07037
B_-3	31.3093	36.4894	0.356154
C_+1	63.5017	66.1264	0.586791
C_-1	118.583	169.101	1.69761
C_-3	63.5579	65.0881	0.590349
D_+0	34.1921	51.7206	0.495521
D_+2	45.4834	46.3077	0.407491
D_-2	34.7952	50.8104	0.492456
E_+0	46.442	64.0919	0.644065
E_+2	43.4201	44.4576	0.372919
E_-2	45.304	56.5035	0.525467
E_-4	45.7779	62.6794	0.530942

Interpretation and limitations

- Independent Yukawa fits are flexible: 18 real Yukawa parameters are fitted to five oscillation observables, so a small χ^2 is not an independent prediction.

- The zero-extra-scalar-coupling boundary is a restricted leading-order approximation, not the complete scalar potential.
- Additional B/C quartics generated by one-loop RGEs are not inserted into the leading fixed-order C_5 coefficient.
- Four extreme-hypercharge models use nonuniform legacy C_5 sources. Full Matchete/RGBeta normalisation and general sequential Wilson matching have not been certified.
- Heavy scalar diagnostics from running UV do not justify using the nominal 1 TeV scalar matching as a physical threshold.

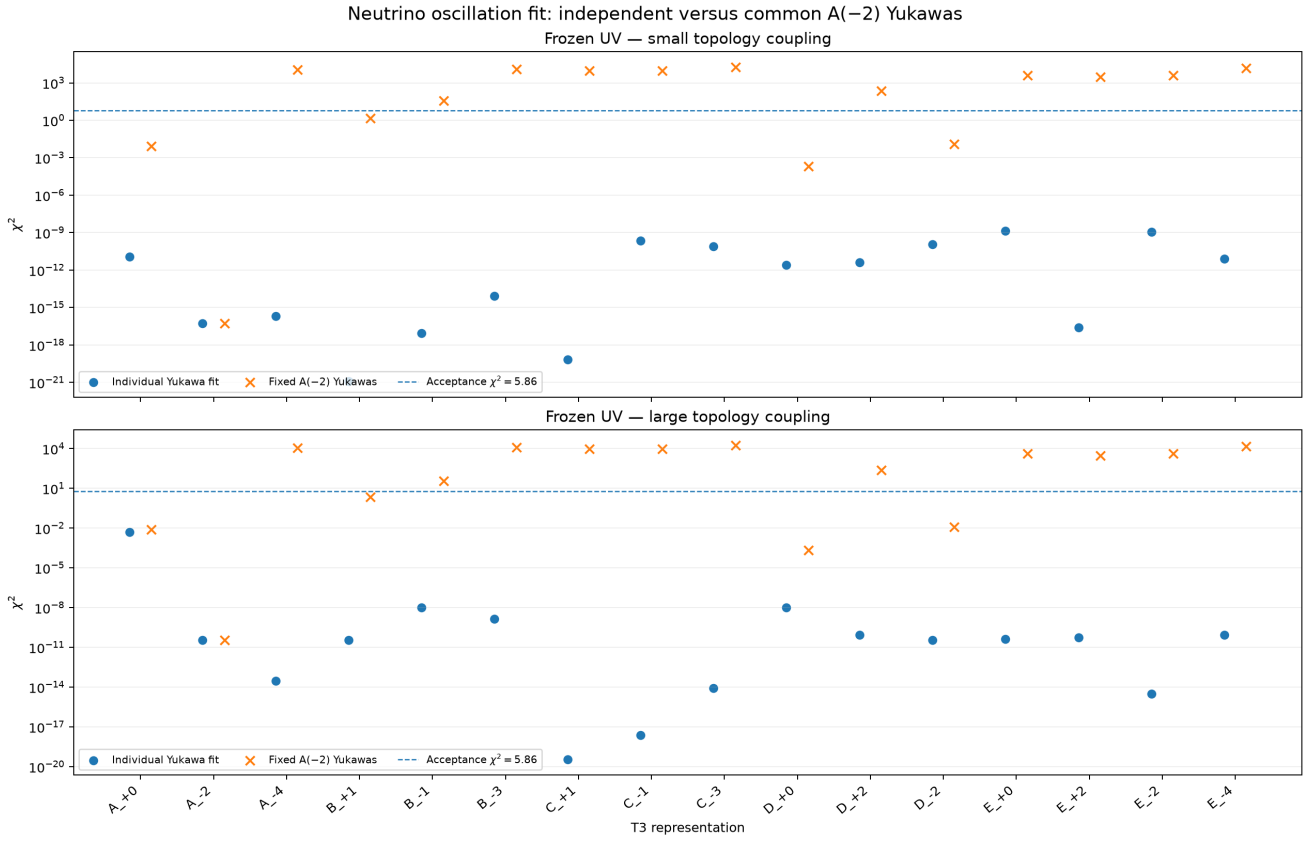


Figure 1: Frozen-UV independent versus shared fits

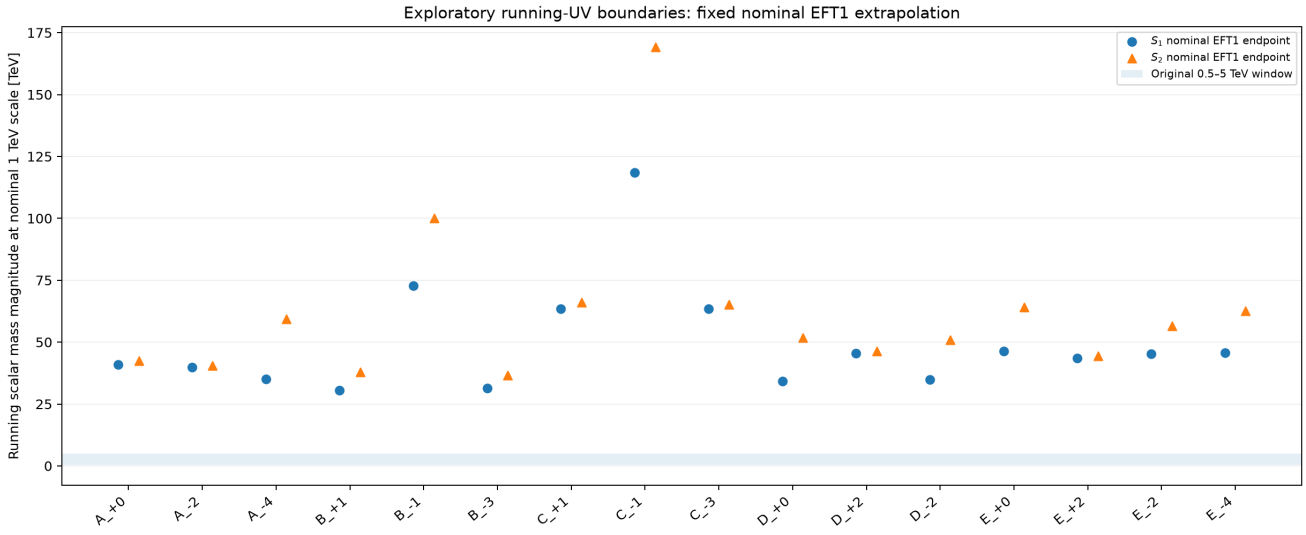


Figure 2: Formal scalar masses at nominal 1 TeV

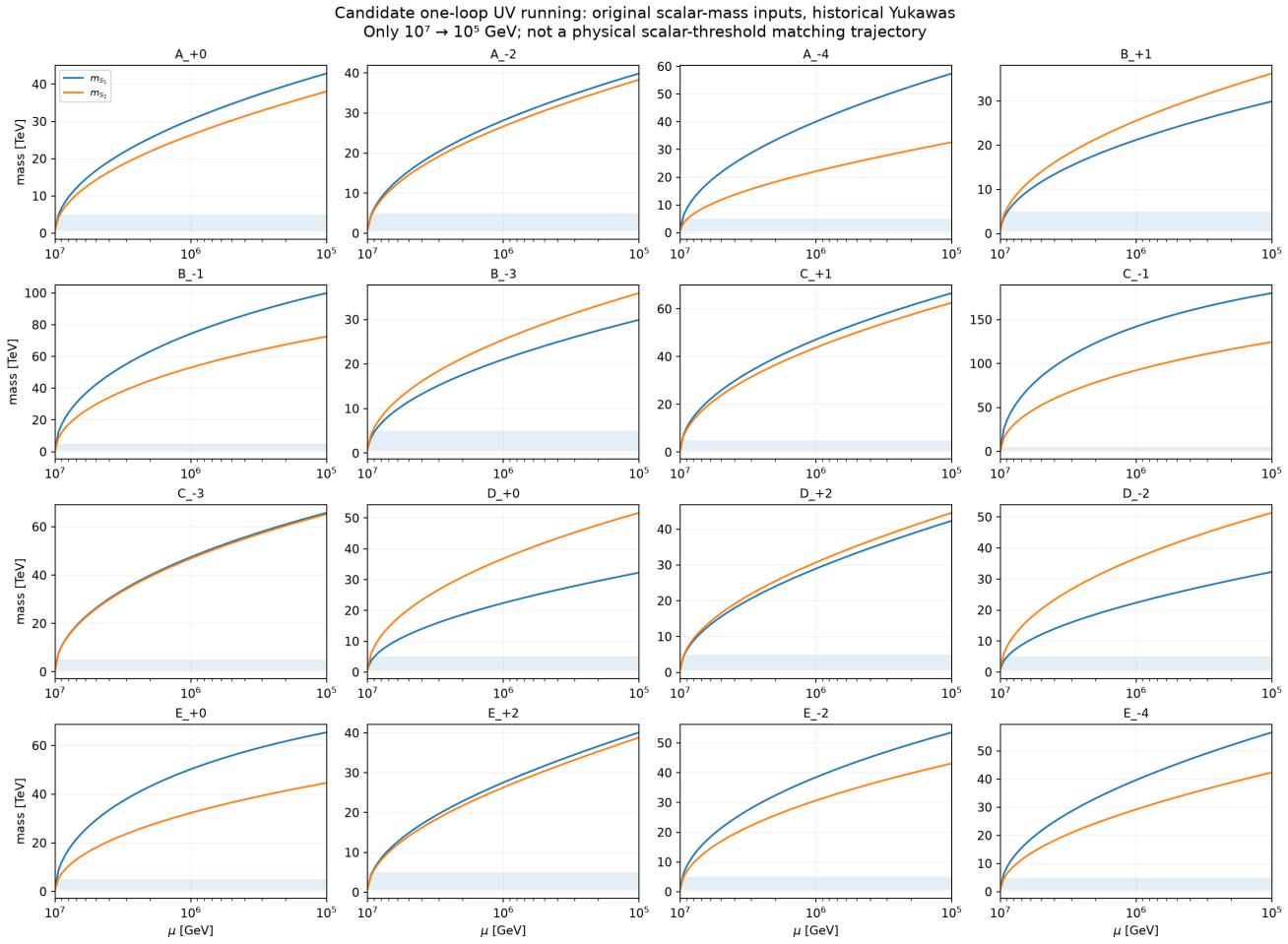


Figure 3: UV-running scalar trajectories, 10^7–10^5 GeV